AI & Computer Vision Knowledge Assistant

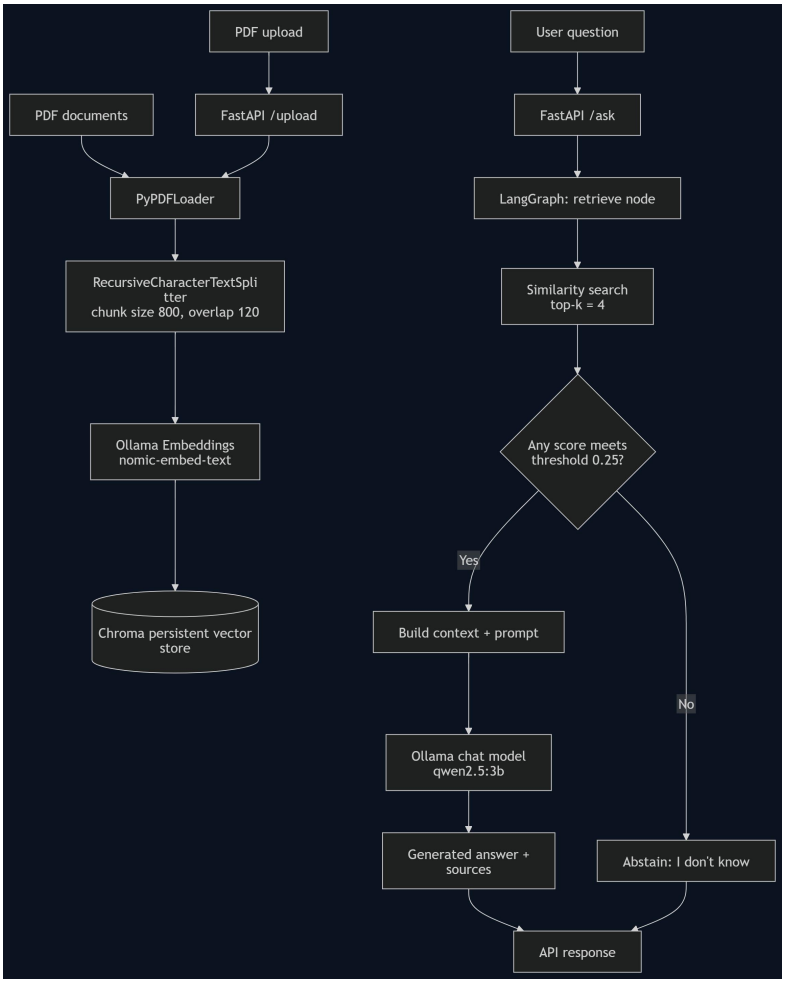
RAG Evaluation Report

LangChain · LangGraph · Chroma · Ollama · FastAPI

1. Objective and System Overview

A Retrieval-Augmented Generation (RAG) assistant ("VisionMind") over AI and Computer Vision PDFs. Uploaded or pre-loaded PDFs are chunked, embedded and stored in Chroma. At question time the top matching chunks are retrieved and given to a local LLM as context; if no chunk is relevant enough the assistant abstains. Every answer returns source filename and page metadata so it can be checked.

Architecture / Workflow (left: indexing; right: question answering)



Configuration

Setting	Value
Chunk size	800 chars
Chunk overlap	120 chars
Top-k	4
Relevance threshold	0.25
Embeddings	nomic-embed-text
Chat model	qwen2.5:3b

Note: *values are initial choices, not proven optimal; the 0.25 threshold is a heuristic that the evaluation below is meant to check.*

Main components

- **PyPDFLoader** extracts page text.
- **RecursiveCharacterTextSplitter** makes chunks.
- **OllamaEmbeddings** turns chunks into vectors.
- **Chroma** stores them persistently.
- **LangGraph** runs retrieve → decide → generate / abstain.
- **FastAPI** serves /upload and /ask.

2. Evaluation Protocol and Results

Protocol. Nine questions from one live session with the assistant were scored by hand against the answer text and the sources shown by the interface: direct factual (1–2), possibly out-of-document (3–4, 7), vague or invalid input (5, 8, 9) and an uploaded-PDF topic (6). One PDF upload was also checked. Scores below come from the recorded chat, not from estimates.

Evaluation Table

ID	Question	Expected key facts	Actual result	Correct?	Source/page correct?	Abstention correct?
1	What is object detection?	Finding instances of known object classes in an image; reported as location / bounding box.	Correct definition with location, scale and bounding-box detail.	Yes	Yes – object_detection.pdf p.1, p.10	n/a
2	What do convolutional layers do?	Apply filters (feature detectors) to the input to detect features.	Correct: filter = feature detector, 3×3 weights, called convolution.	Yes	Yes – deep_learning.pdf p.72–73	n/a
3	Difference between training and test data?	Training data fits the model, test data measures it (general ML; not yet checked in PDFs).	Abstained, but also rambled; retrieved data-augmentation pages.	Verify	No – augmentation pages	Verify
4	What is non-maximum suppression?	Removes duplicate overlapping detections (not yet checked in PDFs).	"I don't know based on the provided documents."	Verify	No – pooling / RNN pages	Verify
5	summarize	Vague: should ask for a topic or abstain.	Incoherent mix of CV techniques and GRU formulas; admits it cannot summarize.	No	No – unrelated chunks	No
6	What's ESP32?	Microcontroller used in IoT Lab 1 (Arduino IDE, Wokwi, LED sketch).	Fluent answer with unsupported details ("ARM-based CPU", "60-node BLE radio").	No	Partial – right PDF (p.2–4), claims not in cited text	n/a
7	What is the capital of Egypt?	Out of scope → abstain.	"I don't know based on the provided documents."	Yes	No – showed résumé / CV pages	Yes
8	..	Invalid input: reject or ask for a question.	Answered about histogram equalization, then said it lacked information.	No	No – p.18 is a title only	No
9	Tell me how you work	Question about the system, not in the PDFs → abstain or explain pipeline.	Described how a Perceptron works; résumé listed as a source.	No	No – Perceptron pages, not the system	No

Reading the table: *"Verify" = the assistant abstained, but whether the PDFs actually contain the answer still has to be checked by hand; expected facts for rows 3–4 are general knowledge, not yet confirmed in the PDFs. 9 of the recommended 10–15 questions are covered.*

Summary

Total questions tested	9 (target 10–15)	**Correct answers**	3 / 9 (Q1, Q2, Q7); 4 wrong, 2 to verify
Correct source/page matches	2 / 6 answered questions	**Correct abstentions**	1 / 3 clear out-of-scope or invalid inputs (Q7 yes; Q8, Q9 no)
Average response time	3-4 seconds	**API/upload checks passed**	Upload 1 / 1 (11 chunks indexed); API validation not tested

3. Analysis, Limitations and Conclusion

Observed Strengths

- Direct, in-document questions (Q1, Q2) were answered correctly with matching filename and page.
- Source metadata made problems easy to spot, e.g. the unsupported ESP32 details and the résumé appearing as a source.
- The abstention branch works for clearly unrelated questions (Q7) and some unanswered ones (Q4); the index is persistent and a PDF upload indexed in one call (11 chunks).

What the Evaluation Showed

On the tested set, 3 out of 9 answers were judged correct, 4 were wrong and 2 abstentions still need a PDF check. Source/page metadata supported the answer in 2 out of 6 answered cases. The system correctly abstained on 1 out of 3 out-of-scope or invalid inputs.

Most common observed issue: the 0.25 threshold lets weak matches through, so vague, empty or meta inputs (Q5, Q8, Q9) get an answer built from loosely related chunks, and sources are shown even when the assistant abstains.

Representative example

- **Question:** "What's ESP32?"
- **Observed behavior:** fluent answer claiming an ARM-based CPU and a "60-node BLE radio"; the cited lab-sheet chunks only mention the ESP32 as the board to program, not these details.
- **Likely cause:** the retrieved chunks were related but did not define the ESP32, and the small qwen2.5:3b model filled the gap from its own memory instead of abstaining.

Limitations and Risks

Only 9 questions, one session and manual scoring, so figures are indicative. Quality depends on PDF text extraction, chunk boundaries, the threshold and the 3B local model. A personal résumé PDF is in the index and surfaces as a source, so the document set needs cleaning. Source lists help auditing but do not prove every sentence is supported. Ollama runs locally and is not reachable from the public internet.

Next Improvements

1. Tune the relevance threshold and top-k on these same questions; add a minimum query length so ".." and "summarize" are rejected.
2. Tell the model to answer only from quoted context and abstain otherwise; hide sources when it abstains.
3. Remove non-course PDFs (résumé) from the index; check Q3 and Q4 against the PDFs.
4. Grow the test set to 10–15 questions, record response times, and add API authentication before public hosting.

Conclusion

The end-to-end pipeline (PDF ingestion, persistent Chroma retrieval, local generation, API) works and answers direct in-document questions correctly with usable sources. It is unreliable on vague, invalid or system-related inputs because the threshold is too permissive. **The main next step is** to tune the threshold and add an input guard, then re-run the full 10–15 question evaluation.